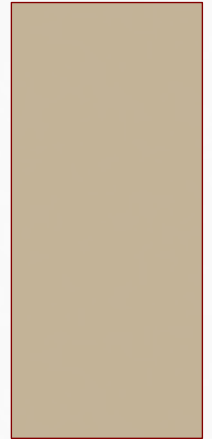


GAME DESIGN PRINCIPLES PART II

ADVANCED GAME DESIGN & DEVELOPMENT

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GAME DESIGN PRINCIPLES

- Most of these principles are common sense
- So why are we going over them?
 - 1 – People (game designers) don't always do what is common sense. You would think they do because it's common sense, but you'd be surprised how often they don't.
 - 2 – It's important to have specific reasons (and research) behind the decisions you're making for your game.

FITTS' LAW

- The tradeoff between speed and accuracy of aimed human muscle movement is referred to as *Fitts' Law*.
- The faster someone moves to a point at a target, the less accurate that movement becomes.
- Question - If core gameplay involves both speed and accuracy...what should the designer make sure they provide the player with?
 - Answer - Clear, easy, and responsive controls

FUNDAMENTAL ATTRIBUTION ERROR

- The willingness of someone to assign situational reasons for their own behavior.
- Your opponent loses and says, “I was lagging...”
- You say, “Stop whining...”
- But the next time you lose, you come up with an excuse like, “My controller wasn’t working...”
- How does this relate to game design?
- If you get negative feedback from your game (during release or playtesting), don’t write it off as the person being dumb or unable to understand how awesome your game really is. Think about what they’re saying and take it seriously.

PACING

- The best games (and all forms of entertainment for that matter) give the audience breathing room between thrills.
- There needs to be areas of sanctuary where the audience is lulled into a sense of security during a point and has time to breathe before they are thrust into the action once more.
- What games did a good job at pacing?
- What games did a poor job at pacing?

SATISFICING VS. OPTIMIZING

- Economic theory says people make decisions through a process (conscious or unconscious)
 - 1. Make a list of all possible outcomes.
 - 2. Assign each possible outcome a probability of happening.
 - 3. Assign each possible choice a value based on its outcome
 - 4. Then choose the optimal outcome
- If a banana split costs \$1 and provides \$3 of happiness, and a sundae costs \$2 and provides \$2 of happiness, which would you choose?

SATISFICING VS. OPTIMIZING

- The problem is that humans don't make decisions like that...it's too time-consuming, complicated, or inconvenient to gather all the information and appropriately optimize every variable.
- Think about it...how do you select what you wore today? Did you spend time trying to optimize your outfit to the weather, your activities, whom you might encounter, etc?
- What people actually do is called *satisficing*. They pick the satisfactory result based on essential rules of thumb that get them close enough to the optimal solution.

SATISFICING VS. OPTIMIZING

- Relating this to game design:
 - Players will approach puzzles and challenges in games using satisficing to find the fastest, easiest solution, rather than optimizing at great length. (Looking up a walkthrough video rather than solving it by yourself.)
 - In games involving multiplayer cooperation, players often choose to satisfice rather than optimize their personal outcome if it will ultimately work to improve the chances of achieving the shared goal. Think about raids...
 - What other examples of satisficing can you think of in games?